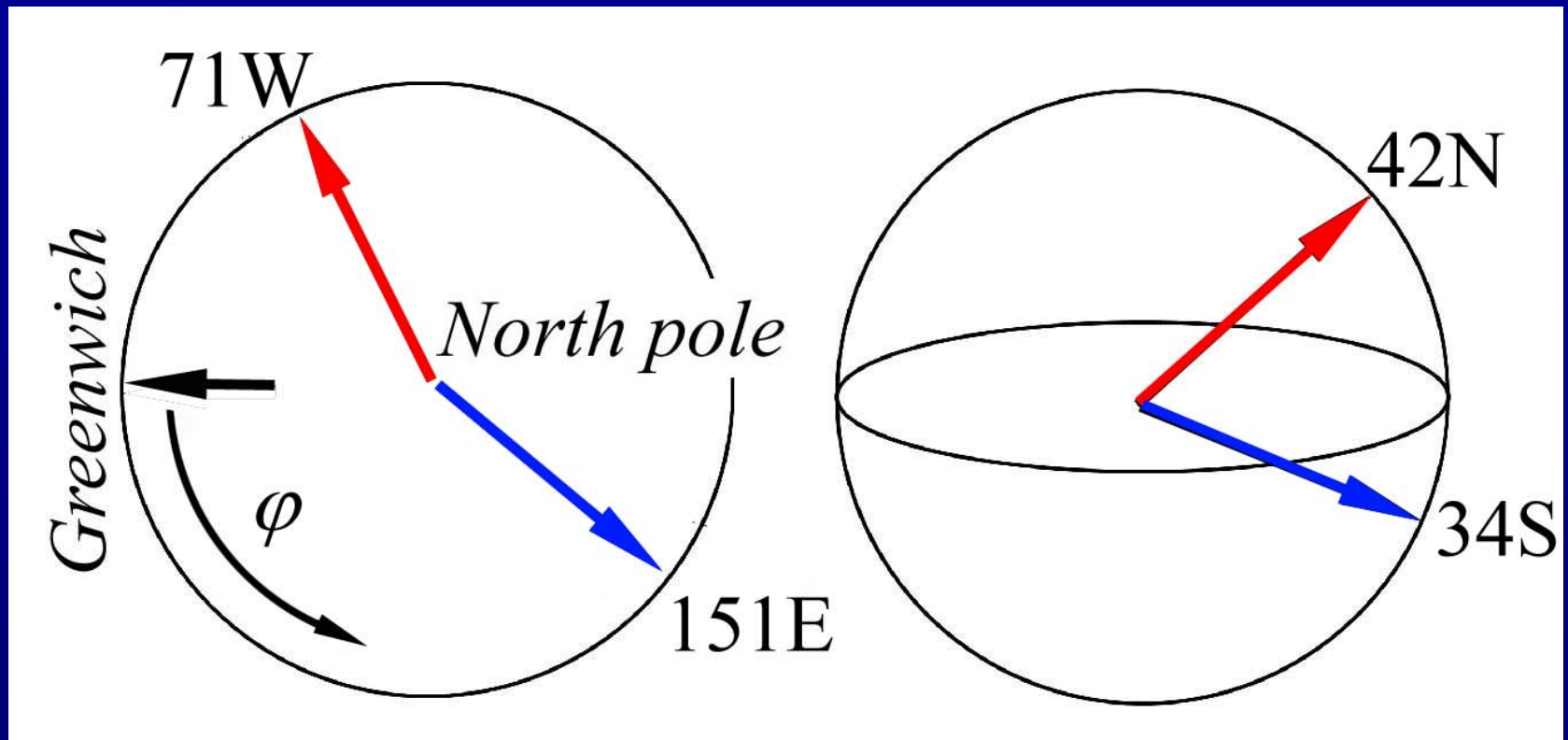


Problem 1.21

Greenwich as x -axis, longitude is φ , (north or south) latitude is $\lambda = 90^\circ - \theta$; or $\theta - 90^\circ$



In spherical system the position vectors are

$$\mathbf{S} = (R, 90 + 34 = 124, 151);$$

$$\mathbf{B} = (R, 90 - 42 = 48, -71 \text{ or } 289)$$

Displacement vector is $\mathbf{S} - \mathbf{B}$.

Convert into Cartesian system

$$\mathbf{e}_r = \cos \theta \mathbf{k} + \sin \theta (\cos \varphi \mathbf{i} + \sin \varphi \mathbf{j})$$

Cosine of the angle included is

$$\begin{aligned}\cos \alpha &= \mathbf{S} \cdot \mathbf{B} / R^2 \\ &= \sin(124)\sin(48)[\cos(151)\cos(-71) + \sin(151)\sin(-71)] \\ &\quad + \cos(124)\cos(48) \\ &= -0.832\end{aligned}$$

$$\arccos(-0.832) = 146^\circ \times \frac{\pi}{180^\circ} = 2.55$$

“Straight line” distance is arc!

$$arc = 6.370 \times 10^3 \text{ km} \times 2.55 = 1.62 \times 10^4 \text{ km}$$

Problem 2.2

not assigned

$$\begin{cases} h = \frac{1}{2}gt^2 \\ \frac{1}{2}h = \frac{1}{2}g(t-1)^2 \end{cases} \longrightarrow \begin{cases} \frac{1}{2}t^2 = (t-1)^2 \\ t = 2 \pm \sqrt{2} \text{ s} \end{cases}$$

For $t = 2 + \sqrt{2} \text{ s}$, $h = 57 \text{ m}$

$2 - \sqrt{2} < 1 \text{ s}$, The 'last second' mentioned in the problem is not available.

Problem 2.3

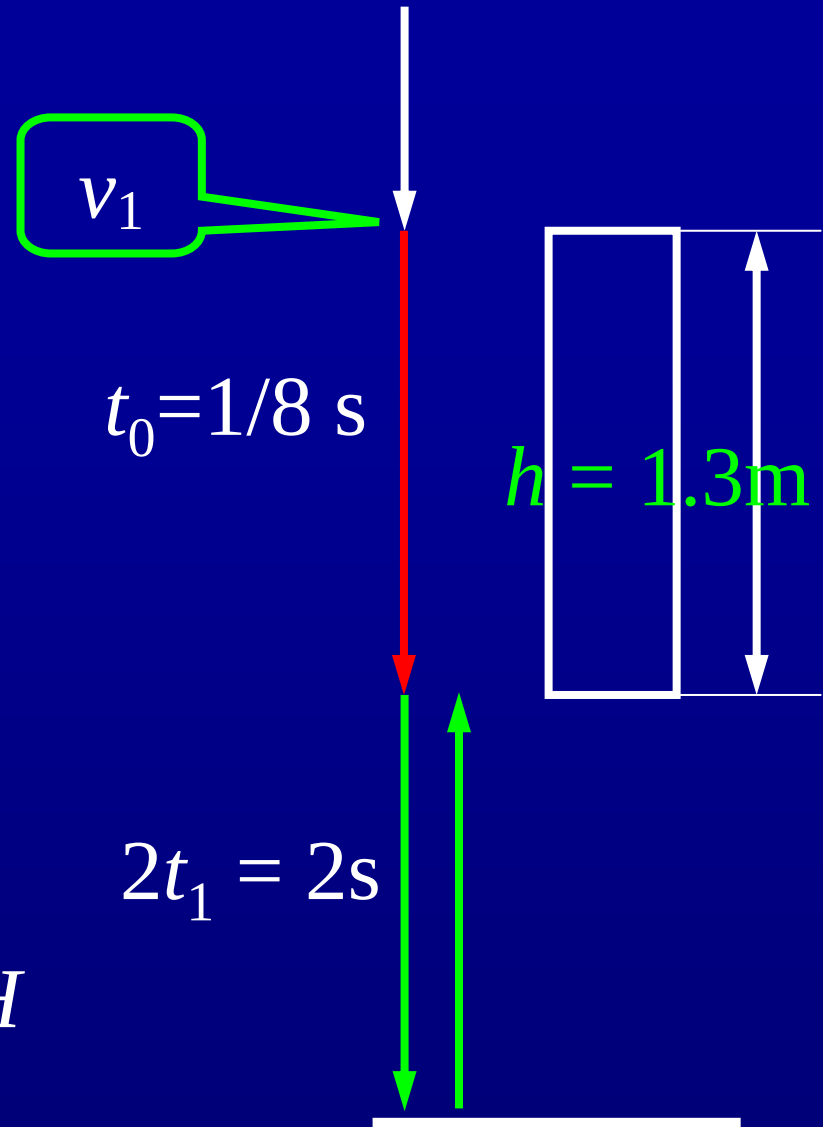
$$H = \frac{1}{2}gt^2$$

$$v_1 = g\left(t - t_0 - \frac{1}{2}t_1\right)$$

$$h = v_1t_0 + \frac{1}{2}gt_0^2$$

$$h, t_0 \text{ — } v_1 ; t_0, t_1 \text{ — } t \text{ — } H$$

$$t = 2.12\text{s}, \quad H = 22.1\text{m}$$



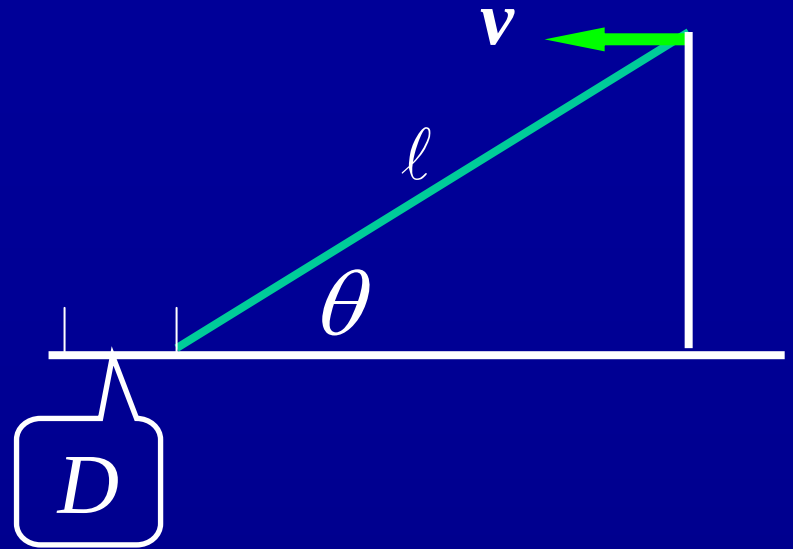
Problem 2.5

$$\ell \sin \theta - \frac{1}{2} g t^2 = 0$$

$$D = vt - \ell \cos \theta$$

$$= v \left(\frac{2\ell \sin \theta}{g} \right)^{1/2} - \ell \cos \theta$$

$D > 0$ pass, $D = 0$ hit, $D < 0$ before



Problem 2.7

$$a = \frac{v^2}{r}$$

$$= \frac{(2.18 \times 10^6)^2}{5.29 \times 10^{-11}} \text{ m/s}^2$$

$$= 8.98 \times 10^{22} \text{ m/s}^2$$

Problem 2.8

$$a_{\text{rot}} = 3.4 \times 10^{-2} \text{ m/s}^2$$

$$a_{\text{rev}} = 5.9 \times 10^{-3} \text{ m/s}^2$$

Problem 2.9 beat

$$\nu_1 = ? \quad \nu_2 = 384 / \text{s}$$

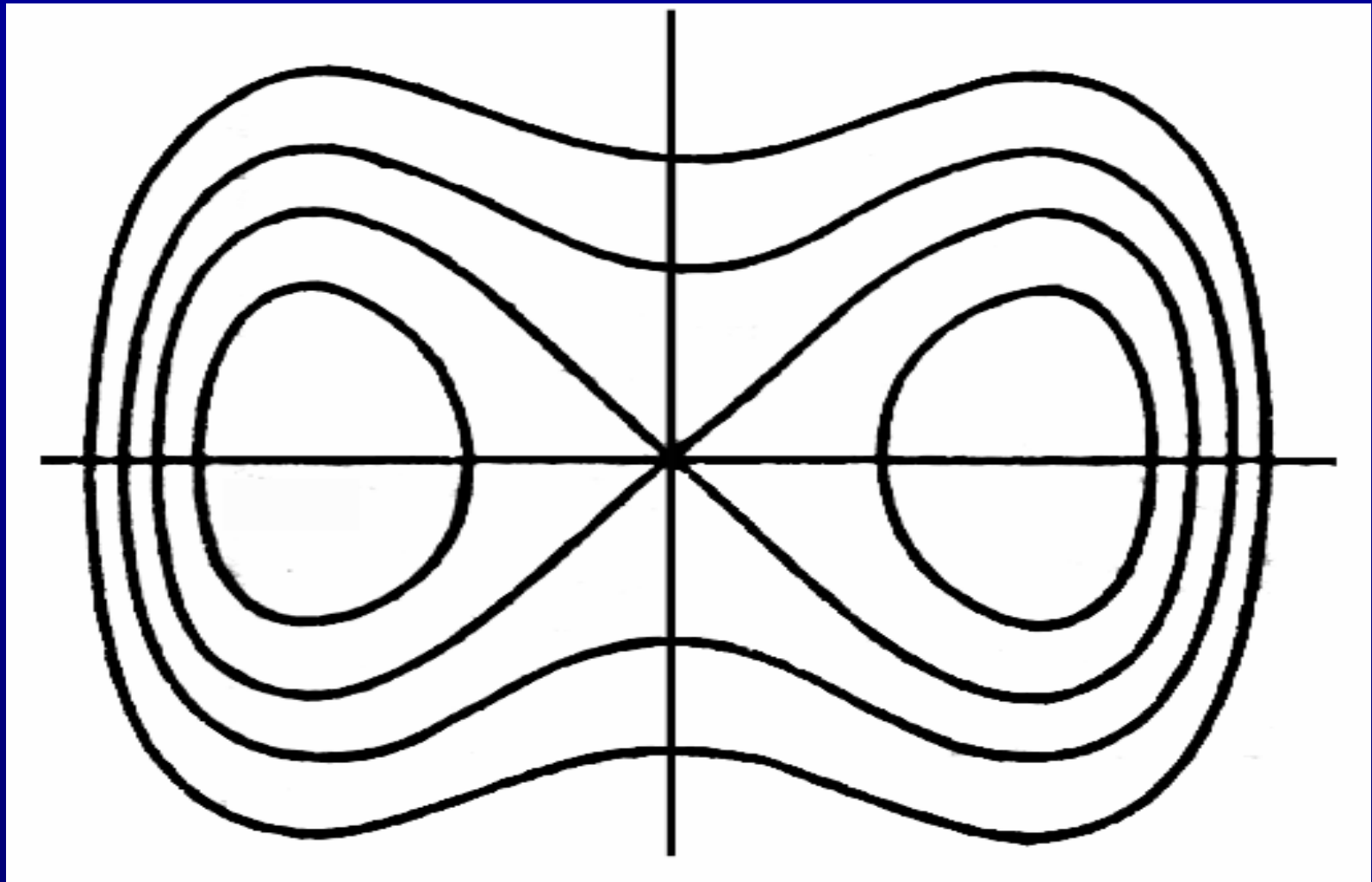
The beat frequency decreases when a small piece of wax is put on a prong of the first fork.

$$\longrightarrow \frac{db}{d\nu_1} > 0$$

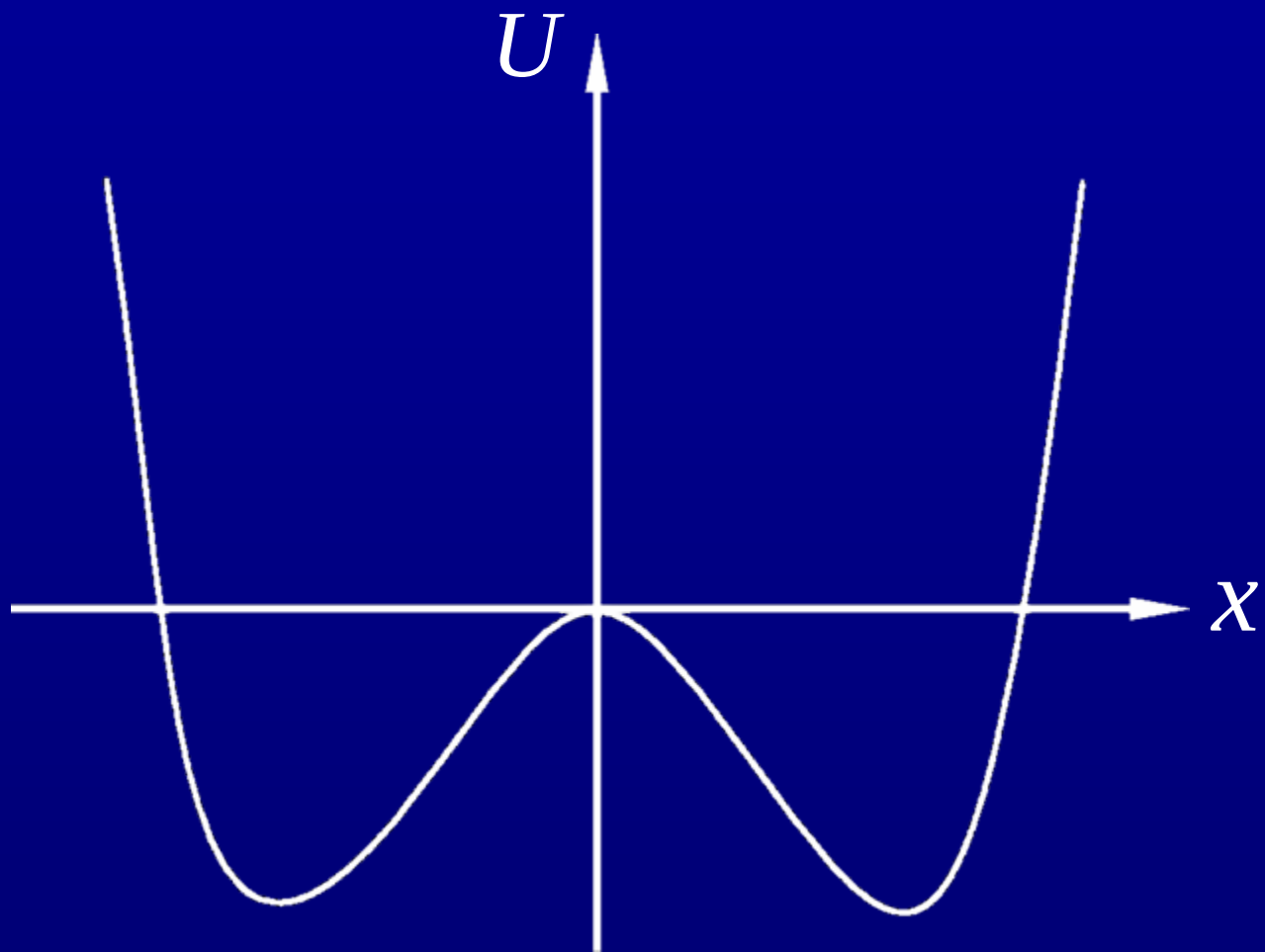
$$\frac{db}{d\nu_1} > 0 \quad \text{means } b = 2 \times \frac{\nu_1 - \nu_2}{2} \text{ not } \nu_2 - \nu_1$$

$$\nu_1 = b + \nu_2 = 387$$

*Problem 2.13



$$\frac{1}{2}m\dot{x}^2 + U(x) = C$$



Problem 2.14 $u(\text{air})$

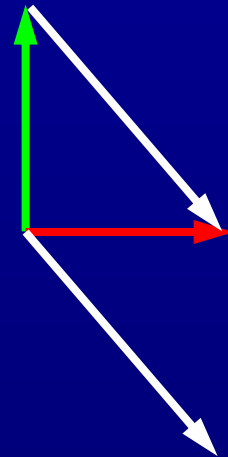
a) $u = 0, v = v', t_0 = 2\ell/v$

b) u due east $v = v' \pm u, \quad t = \frac{\ell}{v' + u} + \frac{\ell}{v' - u}$

c) u due north, v due east

$$v = \sqrt{v'^2 - u^2}$$

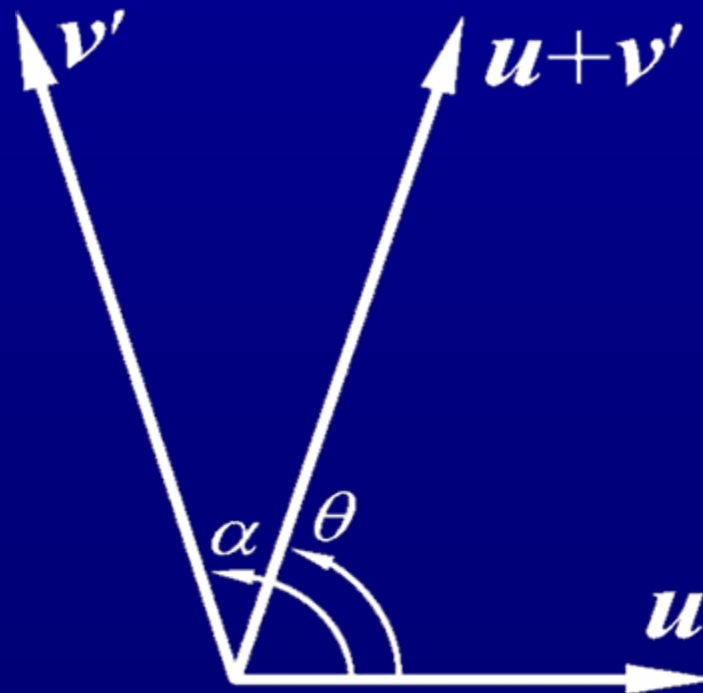
$$t = 2 \frac{\ell}{\sqrt{v'^2 - u^2}} = \frac{2t_0}{\sqrt{1 - \left(\frac{u}{v'}\right)^2}}$$
$$v' > u$$



Problem 2.15 Galilean transformation

Known $\ell = 500\text{m}$, $v' = 3000\text{m/h}$,
 $u = 2000\text{m/h}$, $v_w = 5000\text{m/h}$,

Set the angle relative to flow α



$$t_1 = \frac{\ell}{v_{\perp}} = \frac{\ell}{v' \sin \alpha}; \quad t_2 = \frac{v_{//} t_1}{v_w} = \frac{(u + v' \cos \alpha) t_1}{v_w}$$

$$t = t_1 + t_2$$

$$= \left(1 + \frac{u + v' \cos \alpha}{v_w} \right) t_1$$

$$\frac{dt}{d\alpha} = 0 \longrightarrow \cos \alpha = -\frac{v'}{v_w + u}$$

$$t \approx 0.21 \text{hr} \quad \alpha = 115.37^\circ \quad \theta = 75.2^\circ$$

Problem 2.16

In tube system, virus moves in a straight line.
 $\mathbf{v}' = \text{const.} \longrightarrow \mathbf{a}' = 0, \quad \sum \mathbf{F} = 0$

Gravity-floating; Coriolis force and N from wall; **centrifugal force** and viscosity

In table system $\mathbf{v} = \mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r} = v' \mathbf{e}_\rho + \omega r \mathbf{e}_\phi$

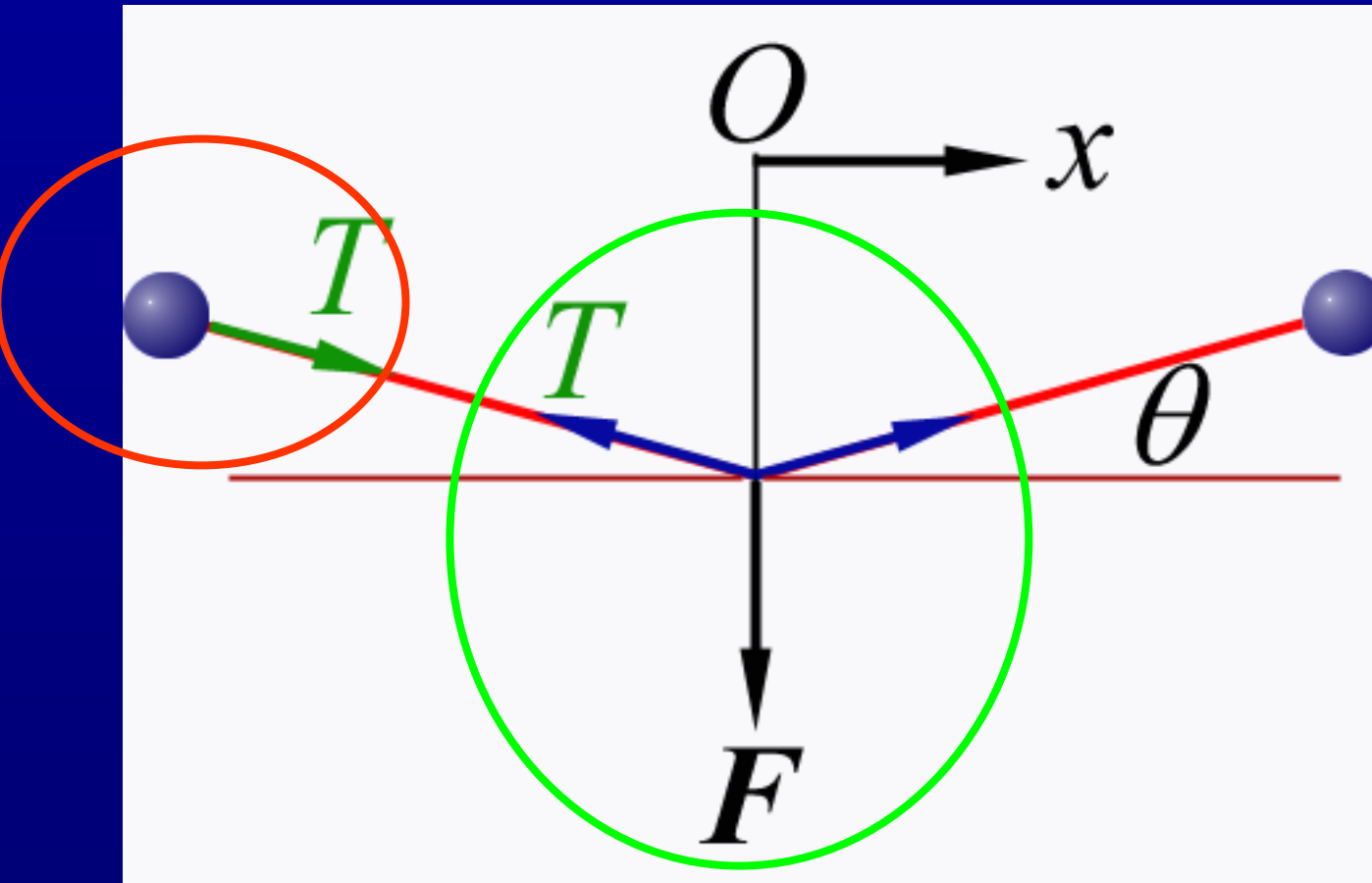
$$\mathbf{a} = \mathbf{a}' + 2\boldsymbol{\omega} \times \mathbf{v}' + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = 2\omega v' \mathbf{e}_\phi - \omega^2 r \mathbf{e}_\rho$$

Coriolis acceleration

centripetal acceleration due to viscosity.

problem 3.1

$$T \cos \theta = ma_x$$
$$F - 2T \sin \theta = 0$$



$$\begin{aligned}
 a_x &= \cos \theta = -\frac{T}{m} \frac{x}{l} \\
 &= -\frac{F}{2m} \frac{1}{\sqrt{1 - \left(\frac{x}{l}\right)^2}} \frac{x}{l} \\
 &= -\frac{F}{2m} \frac{x}{\sqrt{l^2 - x^2}}
 \end{aligned}$$

$$\theta \rightarrow 0, \quad x \rightarrow l, \quad a_x \rightarrow \infty \text{ ?}$$

$$a_x = -\frac{T}{m} \frac{x}{l} \sim T$$

$$T = \frac{F}{2 \sin \theta} \leq T(\text{deformation})$$

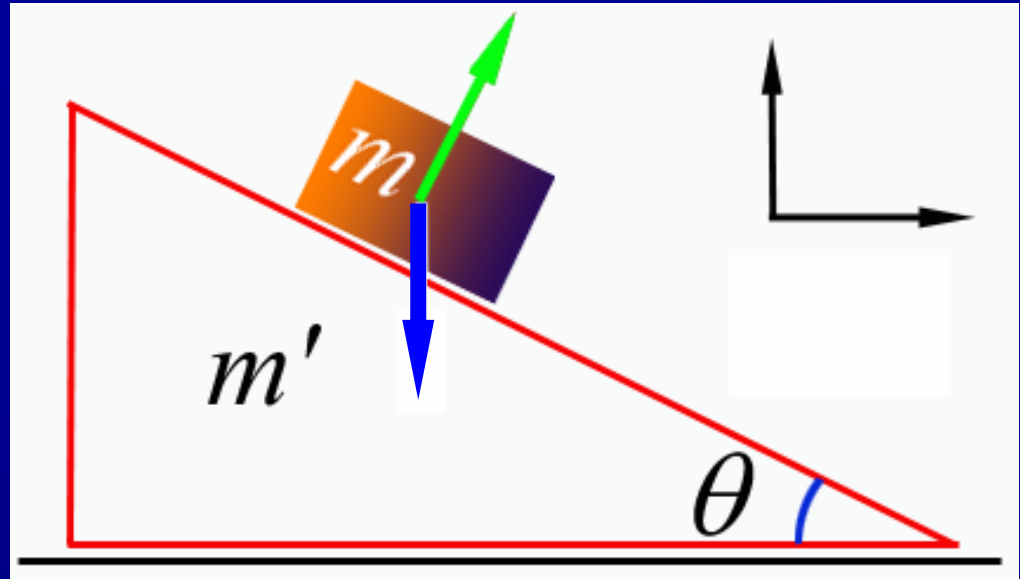
problem 3.2 The condition of stationary is

$$a_{m'x} = a_{mx}$$

a) For m

$$F_n \sin \theta = ma_{mx}$$

$$mg - F_n \cos \theta = 0$$



Eliminating F_n , we have $a_{m'x} = a_{mx} = g \tan \theta$

b) On $m + m'$

$$F = (m' + m)g \tan \theta$$

Eq.(5) $\times \tan \theta$ + Eq.(6)

$$\left[-(m' + m) \tan \theta - m' \cot \theta \right] \ddot{X} - mg = 0$$

So

$$\ddot{X} = -g \frac{m}{m'} \frac{\sin \theta \cos \theta}{1 + \frac{m}{m'} \sin^2 \theta}$$

$$\ddot{x}' = g \frac{m' + m}{m'} \frac{\sin \theta \cos \theta}{1 + \frac{m}{m'} \sin^2 \theta}$$

Discussion: in the table system

$$\ddot{x} = \ddot{x}' + \ddot{X} = g \frac{\sin \theta \cos \theta}{1 + \frac{m}{m'} \sin^2 \theta}$$

$$* \ddot{y} = \ddot{y}' = -\ddot{x}' \tan \theta = -\frac{m' + m}{m'} \ddot{x} \tan \theta$$

It is also a straight line, but with larger slope

$$\left| \frac{\ddot{y}}{\ddot{x}} \right| = \frac{m' + m}{m'} \tan \theta > \tan \theta$$

Eq.(5) $\times \tan \theta$ + Eq.(6)

$$\left[-(m' + m) \tan \theta - m' \cot \theta \right] \ddot{X} - mg = 0$$

So

$$\ddot{X} = -g \frac{m}{m'} \frac{\sin \theta \cos \theta}{1 + \frac{m}{m'} \sin^2 \theta}$$

$$\ddot{x}' = g \frac{m' + m}{m'} \frac{\sin \theta \cos \theta}{1 + \frac{m}{m'} \sin^2 \theta}$$